

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\AR COMPUTER>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.88 nmap-
libidnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\AR COMPUTER>
```

SKILLSQUARK

Network Security Engineering Internship

INTERNSHIP REPORT

Task 1

Network Mapping

Submitted by

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Organization	SkillsQuark
Internship Program	Network Security Engineering Internship
Task	Task 1 – Network Mapping
Tool Used	Nmap 7.99
Submission Date	03 July 2026

1. Objective

The objective of this task is to understand and apply the concept of network mapping using the Nmap network scanning tool. Network mapping is a foundational skill in cybersecurity that allows a security engineer to build a clear picture of the devices, services, and communication paths present within a network. By systematically discovering active hosts, open ports, and running services, an organization can identify its network assets, assess exposure to potential threats, and strengthen its overall security posture.

This task focuses on performing host discovery, TCP port scanning, and UDP port scanning on a local Wi-Fi network using Nmap 7.99, and on interpreting the results from a security perspective. Understanding these fundamentals is essential before moving on to more advanced network security engineering tasks such as vulnerability assessment and penetration testing.

2. Introduction

Network mapping is the process of discovering the devices connected to a network and identifying how they communicate with one another. It provides a visual and logical inventory of hosts, IP addresses, open ports, and the services running on each device. This process is one of the earliest and most important steps in any network security assessment, penetration test, or IT audit, because it establishes visibility into what actually exists on a network before any further analysis can take place.

In cybersecurity, the principle of "you cannot protect what you cannot see" applies directly to network mapping. Unidentified or unmanaged devices connected to a network represent hidden risks, since they may run outdated software, have weak configurations, or expose unnecessary services to potential attackers. By discovering active hosts, scanning for open ports, and identifying the services bound to those ports, a security engineer can:

- Build an accurate and up-to-date inventory of network assets.
- Detect unauthorized or unknown devices connected to the network.
- Identify unnecessary or risky open services that increase the attack surface.
- Prioritize which hosts and services require closer security review.
- Establish a baseline for ongoing network monitoring.

This report documents the tools, environment, commands, and results used to perform network mapping on a local network segment, along with an analysis of the discovered hosts and services.

3. Tools Used

- Nmap 7.99 – an open-source network scanning tool used for host discovery, port scanning, and service detection.
- Windows Command Prompt – used to execute Nmap commands and view scan output.
- Windows Operating System – the host operating system on which Nmap was installed and executed.

4. Environment Setup

The scanning environment was configured as follows before performing the network mapping task:

- A Windows laptop connected to a local Wi-Fi network.
- Nmap 7.99 installed and verified successfully on the system.
- Local IPv4 Address of the scanning machine: 192.168.100.12
- Target network range for scanning: 192.168.100.0/24 (256 addresses)

This environment simulates a typical small office or home network, making it a suitable setting to practice fundamental network mapping techniques before applying them to larger, more complex networks.

5. Commands Used

The following Nmap and Windows commands were executed during this task. Each command is explained below along with its purpose.

5.1 nmap --version

```
C:\> nmap --version
```

This command displays the installed version of Nmap along with the compiled scanning engines and supported libraries. It is typically run first to confirm that Nmap has been installed correctly and to verify which version's features and syntax are available.

5.2 ipconfig

```
C:\> ipconfig
```

This is a native Windows command used to display the current TCP/IP network configuration of the machine, including its IPv4 address, subnet mask, and default gateway. It was used here to identify the scanning machine's own IP address (192.168.100.12) and to determine the correct network range to scan.

5.3 nmap -sn 192.168.100.0/24

```
C:\> nmap -sn 192.168.100.0/24
```

The `-sn` flag performs a "ping scan" or host discovery scan without scanning any ports. This command sends probes to every address in the 192.168.100.0/24 range to determine which hosts are currently online, without attempting to identify open ports or services. It is used to quickly build a list of active devices on the network.

5.4 nmap 192.168.100.12

```
C:\> nmap 192.168.100.12
```

This command performs a default TCP scan against the specified host (in this case, the scanning machine itself). By default, Nmap scans the 1,000 most common TCP ports and reports which ones are open, closed, or filtered, along with the likely service running on each open port.

5.5 nmap -sU 192.168.100.12

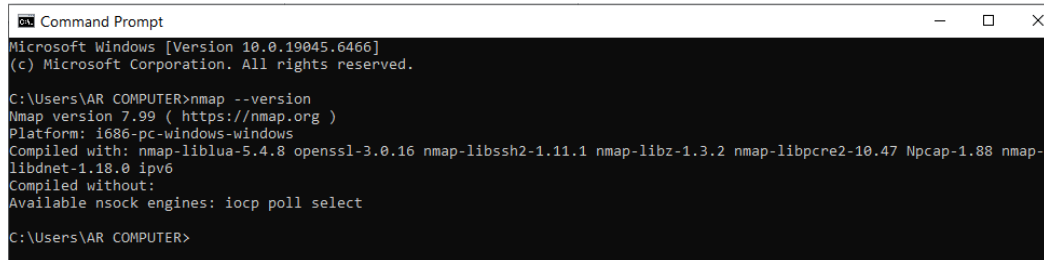
```
C:\> nmap -sU 192.168.100.12
```

The -sU flag instructs Nmap to perform a UDP scan rather than the default TCP scan. UDP scanning is slower and less reliable than TCP scanning because UDP is a connectionless protocol, but it is essential for discovering services such as DNS, DHCP, and NetBIOS that operate over UDP and would otherwise remain undetected.

6. Screenshots

The following screenshots document the practical execution of each command listed above.

Screenshot 1 – Nmap Version



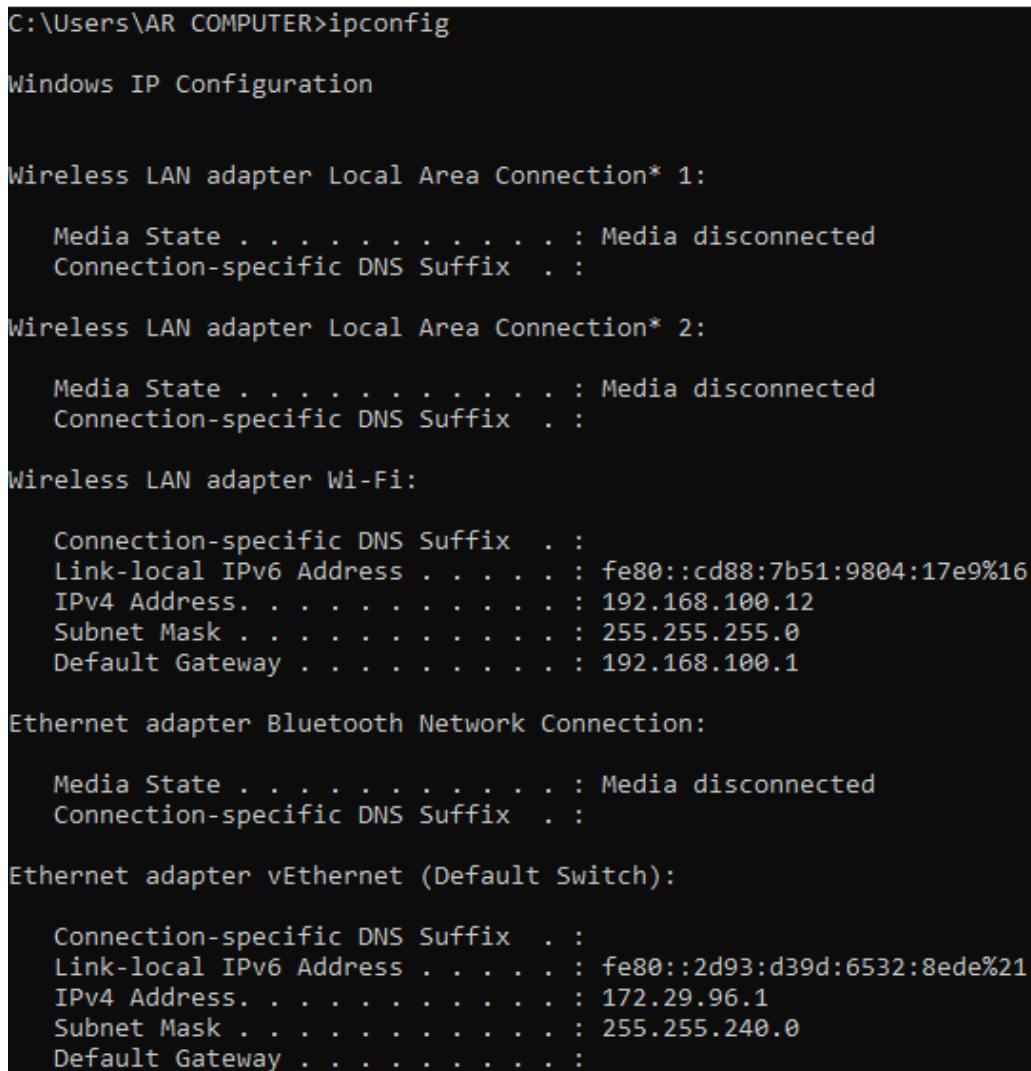
```
Command Prompt
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libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\AR COMPUTER>
```

Figure 1: Output of 'nmap --version' confirming Nmap 7.99 installation.

Screenshot 2 – IP Configuration



```
C:\Users\AR COMPUTER>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::cd88:7b51:9804:17e9%16
    IPv4 Address. . . . . : 192.168.100.12
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2d93:d39d:6532:8ede%21
    IPv4 Address. . . . . : 172.29.96.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :
```

Figure 2: Output of 'ipconfig' showing the local IPv4 address 192.168.100.12.

Screenshot 3 – Network Mapping Scan

```
C:\Users\AR COMPUTER>nmap -sn 192.168.100.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 04:10 -0700
Nmap scan report for 192.168.100.1
Host is up (0.034s latency).
MAC Address: 04:88:5F:45:53:51 (Huawei Technologies)
Nmap scan report for 192.168.100.9
Host is up (1.1s latency).
MAC Address: CA:C7:12:64:A0:3A (Unknown)
Nmap scan report for 192.168.100.30
Host is up (0.86s latency).
MAC Address: DE:40:D8:33:AB:45 (Unknown)
Nmap scan report for 192.168.100.33
Host is up (0.83s latency).
MAC Address: 4E:0E:88:4B:E5:A8 (Unknown)
Nmap scan report for 192.168.100.12
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 17.84 seconds
C:\Users\AR COMPUTER>
```

Figure 3: Output of 'nmap -sn 192.168.100.0/24' host discovery scan.

Screenshot 4 – TCP Port Scan

```
C:\Users\AR COMPUTER>nmap 192.168.100.12
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 04:12 -0700
Nmap scan report for 192.168.100.12
Host is up (0.00072s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2179/tcp  open  vmrpd
3306/tcp  open  mysql
5357/tcp  open  wsapi

Nmap done: 1 IP address (1 host up) scanned in 1.05 seconds
C:\Users\AR COMPUTER>
```

Figure 4: Output of 'nmap 192.168.100.12' TCP port scan.

Screenshot 5 – UDP Port Scan

```
C:\Users\AR COMPUTER>nmap -sU 192.168.100.12
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 04:14 -0700
Nmap scan report for 192.168.100.12
Host is up (0.0011s latency).
Not shown: 992 closed udp ports (port-unreach)
PORT      STATE      SERVICE
137/udp    open|filtered netbios-ns
138/udp    open|filtered netbios-dgm
1900/udp   open|filtered upnp
3702/udp   open|filtered ws-discovery
4500/udp   open|filtered nat-t-ike
5050/udp   open|filtered mmcc
5353/udp   open|filtered zeroconf
5355/udp   open|filtered llmnr

Nmap done: 1 IP address (1 host up) scanned in 50.46 seconds

C:\Users\AR COMPUTER>
```

Figure 5: Output of 'nmap -sU 192.168.100.12' UDP port scan.

7. Asset Inventory Table

The host discovery scan identified the following active devices on the 192.168.100.0/24 network. This inventory serves as a baseline record of known devices for future monitoring and comparison.

Device IP Address	Device Type	Status	MAC Address	Notes
192.168.100.1	Router (Gateway)	Up	Not captured	Huawei router; acts as the default gateway for the network.
192.168.100.9	Unidentified Host	Up	Not captured	Active device on the network; further identification recommended.
192.168.100.12	Laptop (Scanning Host)	Up	Not captured	This device; used to perform all scans in this task.
192.168.100.30	Unidentified Host	Up	Not captured	Active device on the network; further identification recommended.
192.168.100.33	Unidentified Host	Up	Not captured	Active device on the network; further identification recommended.

8. TCP Port Analysis

The TCP scan performed against 192.168.100.12 revealed the following open ports and their associated services:

Port	Service	Description / Security Relevance
135	MSRPC	Microsoft Remote Procedure Call endpoint mapper, used by Windows for inter-process communication. Frequently targeted by malware and worms; should be restricted from untrusted networks.
139	NetBIOS-SSN	NetBIOS Session Service, used for legacy file and printer sharing over NetBIOS. Considered a legacy protocol with known security weaknesses.
445	Microsoft-DS	Used for SMB (Server Message Block) file sharing directly over TCP/IP. Historically exploited by major ransomware and worm attacks (e.g., WannaCry); a high-priority port to secure.
2179	VMRDP	Hyper-V/Virtual Machine Remote Desktop Protocol, used for remote console access to virtual machines running on the host.
3306	MySQL	Default port for the MySQL database service. If exposed without proper authentication and access controls, it can allow unauthorized access to stored data.
5357	WSDAPI	Web Services for Devices API, used by Windows for network device discovery features such as printer and media sharing.

Several of the discovered TCP ports, particularly 135, 139, and 445, are associated with core Windows networking services that are commonly required on local networks for file sharing and management, but are

also frequently targeted in attacks. It is recommended that these ports remain restricted to trusted internal traffic only and are never exposed directly to the internet.

9. UDP Port Analysis

The UDP scan performed against 192.168.100.12 identified the following open or open|filtered ports:

Port	Service	Description / Security Relevance
137	NetBIOS Name Service	Used for NetBIOS name resolution on local networks; a legacy service often left enabled by default.
138	NetBIOS Datagram Service	Used for connectionless communication in NetBIOS-based file and printer sharing.
1900	SSDP	Simple Service Discovery Protocol, used by UPnP-enabled devices to advertise themselves on the network.
3702	WS-Discovery	Used by Windows to automatically discover devices such as printers and scanners on the local network.
4500	IPSec NAT-T	Used for IPSec VPN traffic traversal across NAT (Network Address Translation) devices.
5050	Unassigned / Application-specific	Commonly used by various third-party applications; requires further investigation to confirm the exact service.
5353	mDNS	Multicast DNS, used for local name resolution without a traditional DNS server (e.g., Apple Bonjour, device discovery).
5355	LLMNR	Link-Local Multicast Name Resolution, a legacy name resolution protocol known to be vulnerable to spoofing and relay attacks.

UDP services such as LLMNR (5355) and NetBIOS Name Service (137) are known to be susceptible to spoofing and man-in-the-middle style attacks in enterprise environments, and are often recommended for disabling where legacy compatibility is not required. Discovery-related services such as SSDP and WS-Discovery are generally low risk on trusted local networks but should not be exposed externally.

10. Findings

The complete network mapping exercise produced the following summary of results:

Metric	Result
Total IP addresses scanned	256 (192.168.100.0/24)
Active hosts found	5
Open TCP ports (on 192.168.100.12)	135, 139, 445, 2179, 3306, 5357
Open / Filtered UDP ports (on 192.168.100.12)	137, 138, 1900, 3702, 4500, 5050, 5353, 5355

Out of the full 256-address range defined by the /24 subnet, only 5 hosts responded as active, indicating a small, lightly populated local network consisting of a gateway router, the scanning laptop, and a small number of additional connected devices. The scanning host itself exposed a combination of standard Windows networking ports and one database-related port (3306), which should be reviewed to confirm whether the MySQL service needs to remain active and network-accessible.

11. Conclusion

This task provided practical, hands-on experience with network mapping using Nmap 7.99, one of the most widely used tools in network security engineering. Through host discovery, TCP scanning, and UDP scanning, it was possible to build a clear picture of the devices and services active on a local network segment.

Network mapping is a critical first step in any security assessment because it establishes visibility into what is actually present on a network, including devices, open ports, and running services. Without this visibility, security engineers would have no reliable way to identify unauthorized devices, unnecessary exposed services, or potential entry points for attackers. Tools such as Nmap allow security professionals to continuously audit their networks, detect changes over time, and prioritize remediation efforts based on real, observed risk rather than assumptions.

In a professional network security engineering role, regular network mapping supports asset management, strengthens the organization's security baseline, and forms the foundation for more advanced activities such as vulnerability scanning and penetration testing. This task has reinforced the importance of maintaining ongoing network visibility as a core responsibility of any security engineer.

12. References

1. Nmap Official Website – <https://nmap.org>
2. Official Nmap Documentation and Reference Guide